

Can Large Language Models Predict 3PL Item Parameters in Brazilian Portuguese? An Exploratory Study with ENEM Items

Pedro Otavio Mantovani¹

Advisor: André Carlos Ponce de Leon Ferreira de Carvalho²

University of São Paulo, Institute of Mathematics and Computer Science (ICMC) SP, Brazil

pedro.mantovani6@usp.br¹, andre@icmc.usp.br²

Abstract

3 commercial Large Language Models were used to estimate the **3 logistic parameters** of items from the Brazilian National High School Exam (ENEM). The models could partially explain **item difficulty** (reaching $r = 0.70$), but failed to recover the other two.

Keywords: Large Language Models (LLMs); Item Response Theory (IRT); item parameter estimation.

Background

- Item calibration under **Item Response Theory (IRT)** traditionally requires **pilot samples**, which leads to **high costs and risk of item leakage** (Lang and Tay, 2021). The ENEM itself is scored using the **3PL model** (INEP, 2021a; 2021b).
- Prior work on automated item-parameter prediction has mostly targeted **item difficulty**, using traditional NLP/ML pipelines, almost exclusively for **English-language assessments** (Ferrara et al., 2022; Benedetto, 2023).
- Large Language Models (LLMs) offer a new, low-cost avenue for estimating complete **3-parameter logistic (3PL)** profiles — **discrimination (a)**, **difficulty (b)**, **guessing (c)** — directly from item text. Prior attempts beyond difficulty alone remain scarce, such as neural-network-based 3PL prediction (Belov et al., 2022) and generative simulation of examinee response patterns under a 2PL model (Maeda, 2025), but this remains largely untested outside English and for LLM-based direct estimation.

Objective

To investigate whether state-of-the-art commercial LLMs can estimate complete **3PL item parameters (a, b, c)** directly from the textual content of Brazilian Portuguese assessment items, using a **few-shot prompting** approach.

Method

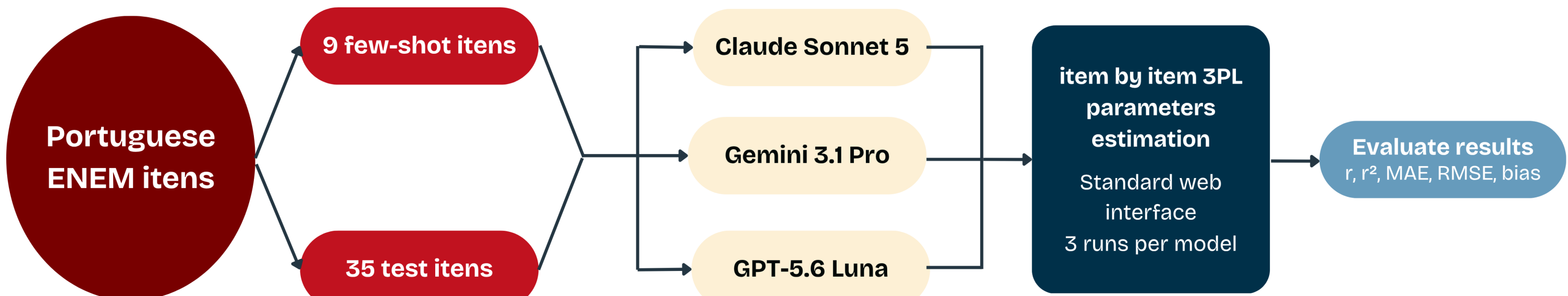


Figure 1: Method scheme

Results

- Difficulty (b)** was the only parameter with a **consistent, above-chance signal**
- Pearson correlation with official values ranged from **0.37 to 0.70** across models for b estimation, which is still insufficient for operational calibration, but is non-random

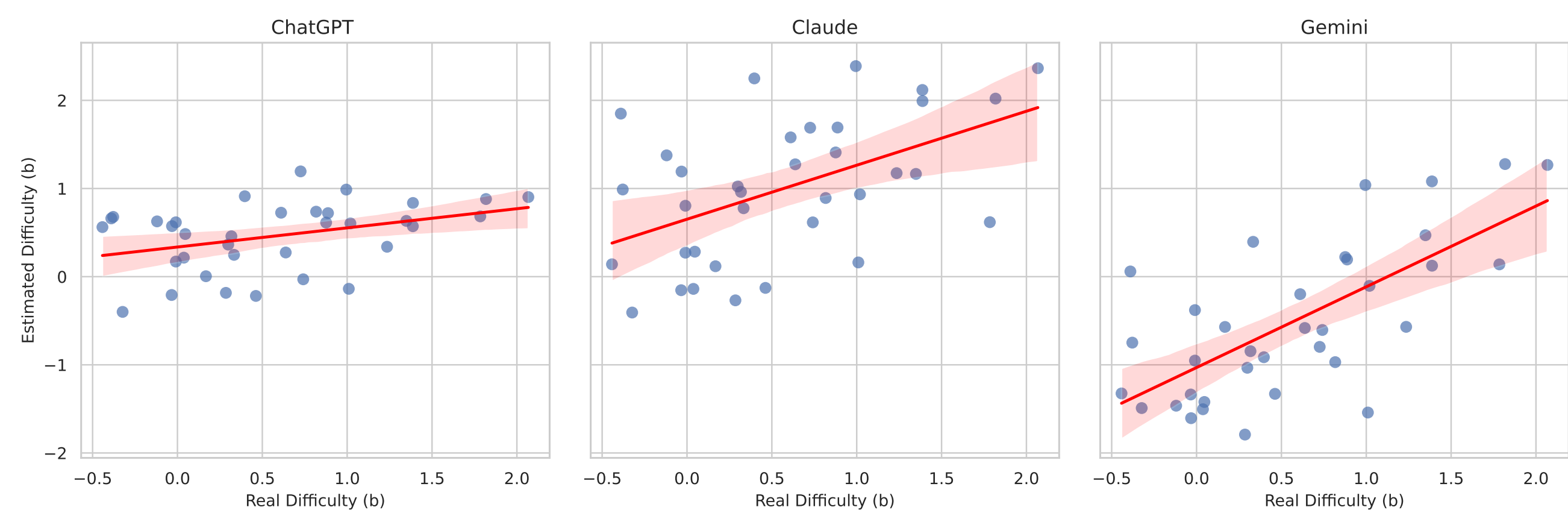


Figure 2: Average model difficulty (b) estimation vs. real parameter.

- Discrimination (a)** and **guessing (c)** showed **near-zero/null predictive power**, with a marked **regression-to-the-mean** pattern

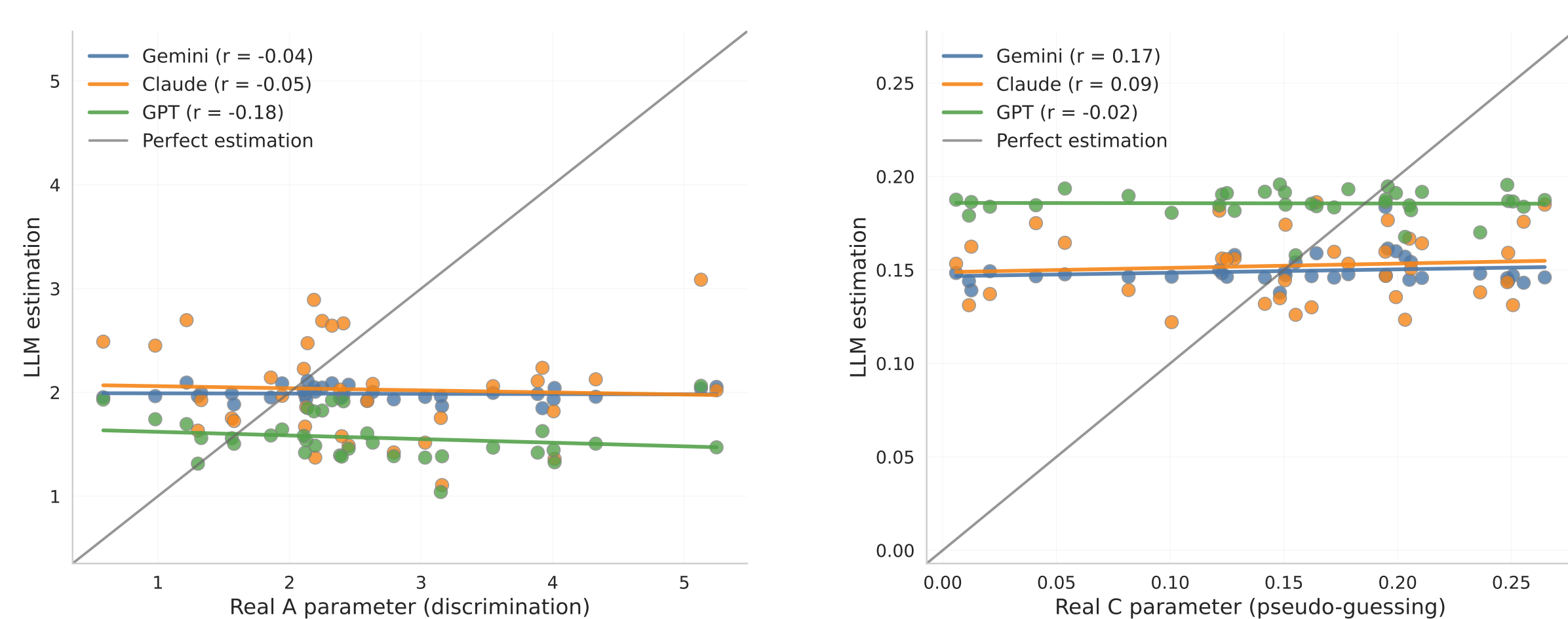


Figure 3: Average model discrimination (a) and pseudo-guessing (c) estimation vs. real parameter.

- Although a and b show essentially no correlation in the official calibration, the LLM-estimated a and b values were **highly intercorrelated (0.33 – 0.94)**, suggesting a shared, possibly difficulty-driven heuristic rather than independent estimation of the two parameters

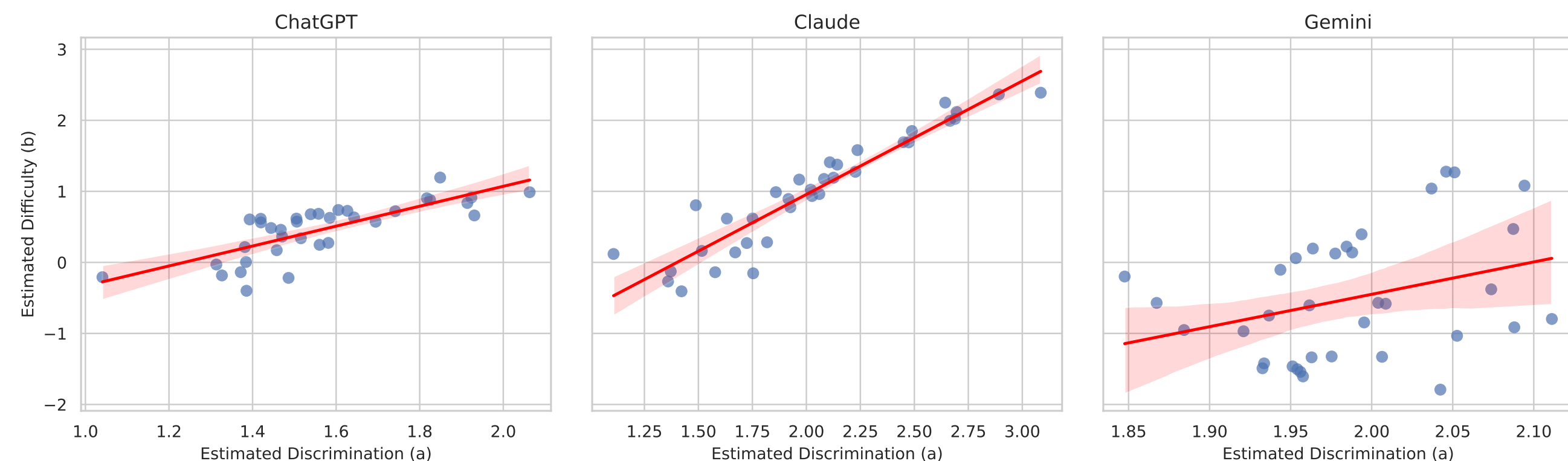


Figure 4: Correlation between estimated a and b parameters.

- Stability and ensembling:** Between-run agreement, measured via Intraclass Correlation Coefficient (ICC), ranged from **0.60 to 0.93** across models. Despite this variability, averaging estimates across the three runs mitigated noise and prevented poor individual generations from degrading results. The resulting **ensemble matched or outperformed the best single run**, without requiring ground-truth data to select which run to trust.

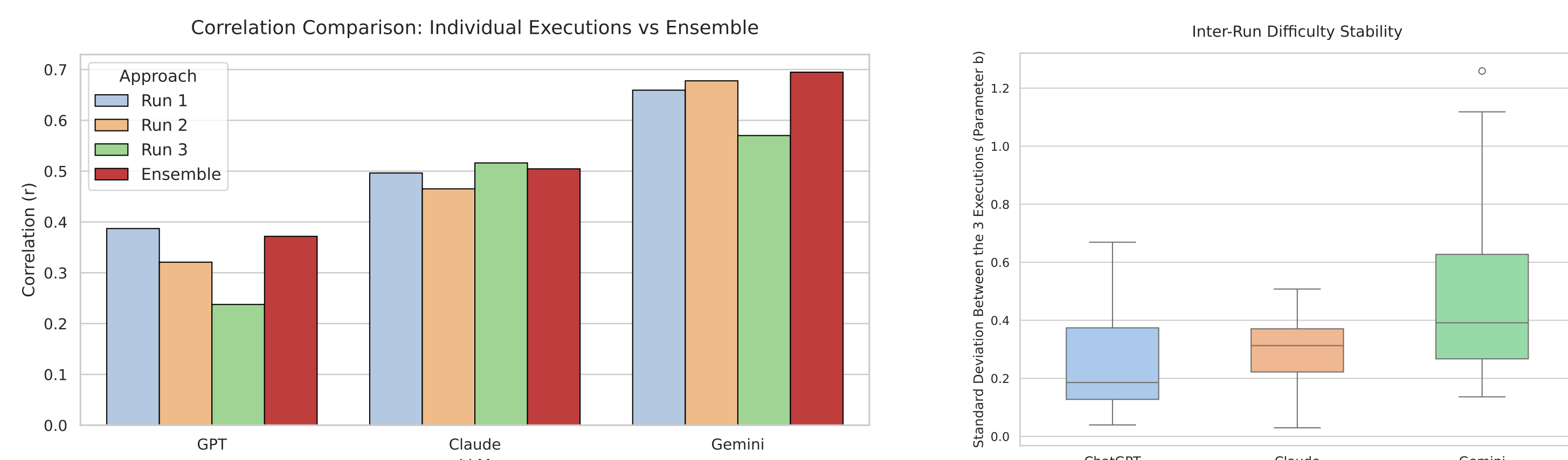


Figure 5: Comparison between r values for single runs and average value.

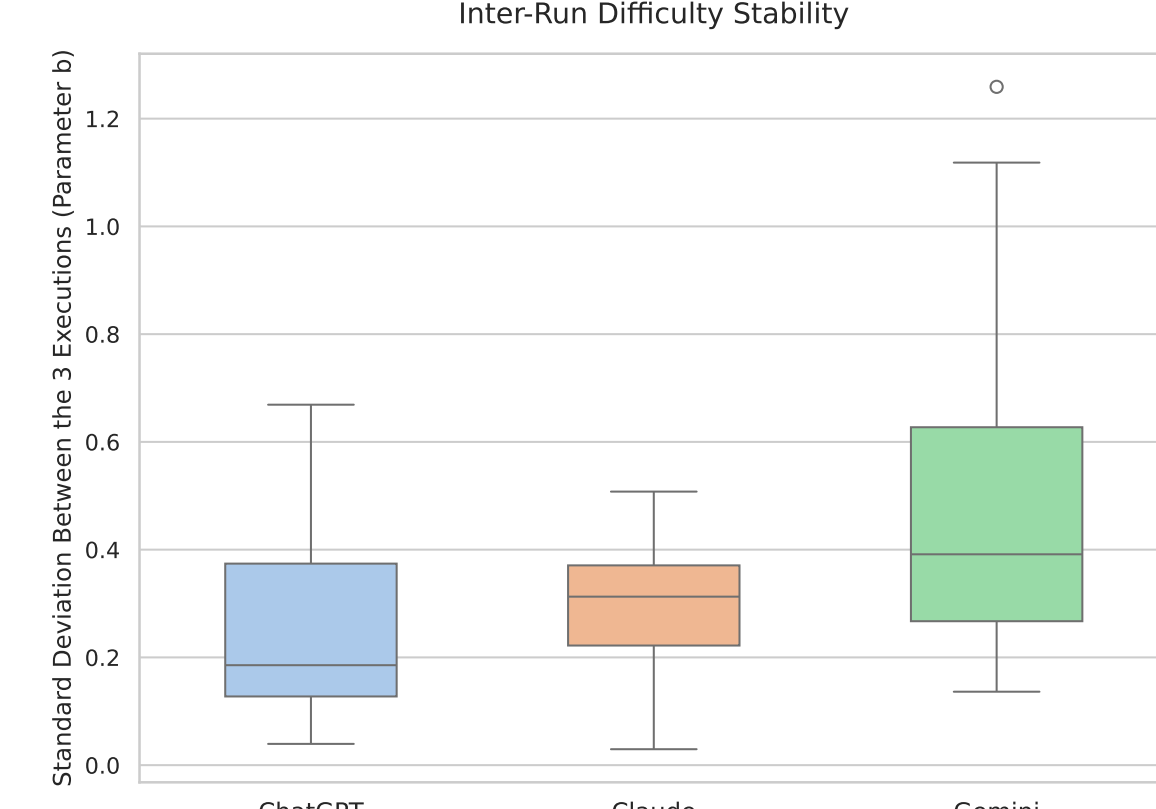


Figure 6: Standard Deviation between the three runs.

Discussion

Despite being an exploratory, small-scale study, using off-the-shelf, non-fine-tuned, general-purpose commercial LLMs, in Portuguese (a comparatively lower-resource language for these systems), with only 9 few-shot examples, the difficulty-prediction performance already **falls within the range reported by prior studies** using dedicated NLP/ML pipelines trained specifically for item-difficulty prediction (Belov et al., 2022; Benedetto, 2023). The comparatively weaker recovery of discrimination echoes previous generative approaches, which also struggled to model this parameter (Maeda, 2025).

This suggests LLMs are already capturing psychometrically relevant signal “for free,” without any task-specific training, a **meaningful baseline for future AI-assisted psychometrics**.

The a, b correlation at estimations gives a plausible explanation for why a shows no predictive power despite b showing a moderate one. This suggests that another approach for estimating a , might have better results.

Limitations & Future Work

- Small test set** (35 items) and few-shot example set (9 items), restricted to a single content domain (language items) → evaluate on larger, calibrated item sets spanning multiple domains (e.g., mathematics, sciences) to disentangle sample-size from domain effects
- Near-zero performance for a and c** , with a marked regression-to-the-mean pattern → test whether larger few-shot sets or fine-tuning reduce this pattern, before concluding these parameters are unrecoverable from text alone
- Unstructured, non-explanatory prompting** → adopt structured, explainable prompting (e.g., asking the model to justify each estimate) to diagnose systematic errors such as the $a-b$ collinearity observed here
- Single language and exam** (Portuguese, ENEM) → compare against English-language items, given LLMs’ far greater exposure to English standardized-test data during pretraining, even though LLMs have shown strong performance on Brazilian entrance-exam items more generally (Godoy, 2025)
- General-purpose, off-the-shelf models only** → assess how much more advanced or specialized models, or LLMs used as features within an interpretable downstream model, close the remaining gap

Conclusion

Contemporary LLMs can already capture some psychometrically relevant information from Brazilian Portuguese assessment content — particularly for **item difficulty** — at levels broadly comparable to specialized NLP/ML approaches, despite an exploratory, small-scale, non-fine-tuned setup. Results are not yet sufficient for operational item calibration — a level of precision still below what is required to meaningfully reduce pilot-sample requirements in practice (Ulitzsch et al., 2026) — but they support continued investigation of LLMs as a **low-cost complement to traditional IRT calibration**.

Financial support: I thank National Council for Scientific and Technological Development (CNPq) and Institute of Mathematics and Computer Science (ICMC) for the financial support.



Full references, prompts, code & results:

github.com/pedro-mantovani/IACAT_ipo